Ex. No.	2	Name	Ajmal Rahman A
Date	18-10-2026	Reg. No.	192324094
Sub. Name	Deep Learning	Sub. Code	CSA4702
Semester	MS2 September 2026		

EXERCISE NAME

COVARIANCE ANALYSIS OF RETAIL CUSTOMER BEHAVIOUR

SCENARIO

A retail company collects the annual income, the annual spending and the purchase frequency of its customers. Analyse the covariance matrix of these three features and determine which of them show a strong positive relationship and which show a strong negative relationship, so that the company can decide which features carry the same information about a customer.

ALGORITHM

Step 1 : Import the packages used in this exercise — NumPy for the matrix algebra, Pandas for the tabular handling of the customer records and Matplotlib for the plots.

Step 2 : Build the data matrix X of order 150 × 3, in which each of the 150 rows is one customer and the three columns are annual income, annual spending and purchase frequency. Generate the spending so that it rises with the income, and the purchase frequency so that it falls as the income rises.

Step 3 : Compute the descriptive statistics — count, mean, standard deviation, minimum, quartiles and maximum — for the three features.

Step 4 : Compute the mean vector $\mu = (1/n)\ \Sigma\ x_i$ and form the mean-centred deviation matrix $Xc = X - \mu$, so that every column of Xc has mean zero and only the variation about the mean is left.

Step 5 : Compute the covariance matrix from first principles as $S = (1/(n - 1)) \times Xc^T \cdot Xc$. The inner dimensions of Xc^T (3 × 150) and Xc (150 × 3) agree, so S is of order 3 × 3; its diagonal holds the variance of each feature and its off-diagonal element s_{ij} holds the covariance of features i and j.

Step 6 : Verify the result against the library computation np.cov(X), which must give the same matrix.

Step 7 : Convert the covariance matrix into the correlation matrix by $r_{ij} = s_{ij} / (s_i \times s_j)$, where s_i is the standard deviation of feature i. This removes the units, so that every relationship is measured on the common scale −1 to +1 and the three pairs can be compared with one another.

Step 8 : Classify every feature pair by the sign of r — positive if the two features rise together, negative if one rises while the other falls — and by the magnitude of r — very strong for $|r| \geq 0.80$, strong for $|r| \geq 0.60$, moderate for $|r| \geq 0.40$, weak for $|r| \geq 0.20$ and negligible below it.

Step 9 : Plot the three pairwise scatter plots with the fitted trend line of each, so that the direction of every relationship can be seen.

Step 10 : Plot the covariance map and the correlation map as heat maps with the values written inside the cells, and report the strongest positive and the strongest negative relationship. Stop.

SOURCE CODE

Module 1 : Importing the Required Packages

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

plt.rcParams["font.family"]          = "serif"
plt.rcParams["font.serif"]           = ["Cambria", "Caladea", "DejaVu Serif"]
plt.rcParams["axes.spines.top"]      = False
plt.rcParams["axes.spines.right"]    = False
plt.rcParams["legend.frameon"]       = False
```

Module 2 : Building the Retail Customer Dataset

```
np.random.seed(42)
N = 150                                          # number of customers

income    = np.clip(np.random.normal(6.5, 1.9, N), 2.2, 13.0)      # lakh / year
spending  = np.clip(0.42*income + np.random.normal(0, 0.28, N),
                    0.4, 6.0)               # lakh / year
frequency = np.clip(44 - 3.1*income + np.random.normal(0, 3.2, N),
                    2, 60)                 # visits / year

df = pd.DataFrame({
    "Customer ID"              : ["C%03d" % (i + 1) for i in range(N)],
    "Annual Income (Rs lakh)"  : np.round(income, 2),
    "Annual Spending (Rs lakh)": np.round(spending, 2),
    "Purchase Frequency (/yr)" : np.round(frequency, 1)})

FEATURES = ["Annual Income (Rs lakh)",
            "Annual Spending (Rs lakh)",
            "Purchase Frequency (/yr)"]
SHORT    = ["Income", "Spending", "Frequency"]

X = df[FEATURES].to_numpy(dtype=float)           # data matrix , shape (150, 3)
print("Data matrix X :", X.shape)
print(df.head(10).to_string(index=False))
```

Module 3 : Descriptive Statistics of the Three Features

```
desc = df[FEATURES].describe().round(3)
print(desc)
```

Module 4 : Mean Vector and the Mean-Centred Deviation Matrix

```
mean_vec = X.mean(axis=0)                        # mean vector , shape (3,)
Xc       = X - mean_vec                     # deviation matrix , shape (150, 3)

print("Mean vector      :", np.round(mean_vec, 3))
print("Column means of Xc :", np.round(Xc.mean(axis=0), 12))
```

Module 5 : Covariance Matrix from First Principles and with NumPy

```
n         = X.shape[0]
cov_manual = (Xc.T @ Xc) / (n - 1)               # S = (1/(n-1)) * Xc^T . Xc
cov_numpy  = np.cov(X, rowvar=False)             # same result from NumPy

print("Covariance matrix (manual) :\n", np.round(cov_manual, 4))
print("Matches np.cov :", np.allclose(cov_manual, cov_numpy))
print("Variances on the diagonal :", np.round(np.diag(cov_manual), 4))
```

Module 6 : Correlation Matrix and Strength of the Relationships

```
sd        = np.sqrt(np.diag(cov_manual))
```

```
corr      = cov_manual / np.outer(sd, sd)         # r_ij = s_ij / (s_i * s_j)

def strength(r):
    a = abs(r)
    return ("Very strong" if a >= 0.80 else "Strong" if a >= 0.60 else
            "Moderate"    if a >= 0.40 else
            "Weak"        if a >= 0.20 else "Negligible")

pairs = []
for i in range(3):
    for j in range(i + 1, 3):
        r = corr[i, j]
        pairs.append((SHORT[i], SHORT[j], cov_manual[i, j], r,
                      "Positive" if r > 0 else "Negative", strength(r)))
        print("%-10s vs %-10s  cov %8.4f  r %+6.3f  %-8s  %s" % pairs[-1])

print("Matches pandas .corr() :",
      np.allclose(corr, df[FEATURES].corr().to_numpy()))
```

Module 7 : Visualisation of the Pairwise Relationships

```
COLORS = ["#2F6DB5", "#C0702A", "#4E8F59"]
fig, axes = plt.subplots(1, 3, figsize=(8.4, 2.9))
for ax, (a, b, cv, r, sign, st), cl in zip(axes, pairs, COLORS):
    xa, xb = X[:, SHORT.index(a)], X[:, SHORT.index(b)]
    ax.scatter(xa, xb, s=12, c=cl, alpha=0.85)
    m, c = np.polyfit(xa, xb, 1)
    xs = np.linspace(xa.min(), xa.max(), 50)
    ax.plot(xs, m*xs + c, color="#404040", lw=1.2, ls="--")
    ax.set_xlabel(a);  ax.set_ylabel(b)
    ax.set_title("%s vs %s   (r = %+.3f)" % (a, b, r), fontsize=9)
    ax.grid(alpha=0.5);  ax.set_axisbelow(True)
fig.tight_layout()
fig.savefig("plot1_scatter.png", dpi=300, bbox_inches="tight")
```

Module 8 : Visualisation of the Covariance and Correlation Maps

```
fig, axes = plt.subplots(1, 2, figsize=(8.0, 3.6))
for ax, M, name, cmap, vlim in [
        (axes[0], cov_manual, "(a) Covariance matrix", "PuOr", None),
        (axes[1], corr,       "(b) Correlation matrix", "RdBu", 1.0)]:
    lim = vlim if vlim else np.abs(M).max()
    im  = ax.imshow(M, cmap=cmap, vmin=-lim, vmax=lim)
    ax.set_xticks(range(3), SHORT, rotation=20, ha="right")
    ax.set_yticks(range(3), SHORT)
    ax.set_title(name, fontsize=10)
    for i in range(3):
        for j in range(3):
            ax.text(j, i, "%.3f" % M[i, j], ha="center", va="center",
                    fontsize=8,
                    color="white" if abs(M[i, j]) > 0.55*lim else "#1A1A1A")
    fig.colorbar(im, ax=ax, fraction=0.046, pad=0.04)
fig.tight_layout()
fig.savefig("plot2_covariance_map.png", dpi=300, bbox_inches="tight")
```

OUTPUT

Table 1 - Retail Customer Dataset (First 10 Customers)

S.No	Customer ID	Annual Income (Rs lakh)	Annual Spending (Rs lakh)	Purchase Frequency (/yr)
1	C001	7.44	3.20	18.3
2	C002	6.24	2.72	22.9
3	C003	7.73	3.06	22.4
4	C004	9.39	4.01	16.8
5	C005	6.06	2.63	25.2
6	C006	6.06	2.34	25.6
7	C007	9.50	4.51	18.6
8	C008	7.96	3.48	17.4
9	C009	5.61	2.02	28.4
10	C010	7.53	3.35	20.0

Table 2 - Descriptive Statistics of the Three Features

Statistic	Annual Income (Rs lakh)	Annual Spending (Rs lakh)	Purchase Frequency (/yr)
Count	150	150	150
Mean	6.348	2.686	24.495
Std. Deviation	1.779	0.794	6.376
Minimum	2.200	0.790	8.300
25th Percentile	5.225	2.142	19.725
Median (50%)	6.395	2.690	24.300
75th Percentile	7.430	3.198	29.150
Maximum	11.180	4.610	44.100

Table 3 - Covariance Matrix S (order 3 × 3)

Feature	Income	Spending	Frequency
Income	3.1645	1.3169	-9.9761
Spending	1.3169	0.6300	-4.1199
Frequency	-9.9761	-4.1199	40.6544

Table 4 - Correlation Matrix R (order 3 × 3)

Feature	Income	Spending	Frequency
Income	+1.0000	+0.9327	-0.8795
Spending	+0.9327	+1.0000	-0.8141
Frequency	-0.8795	-0.8141	+1.0000

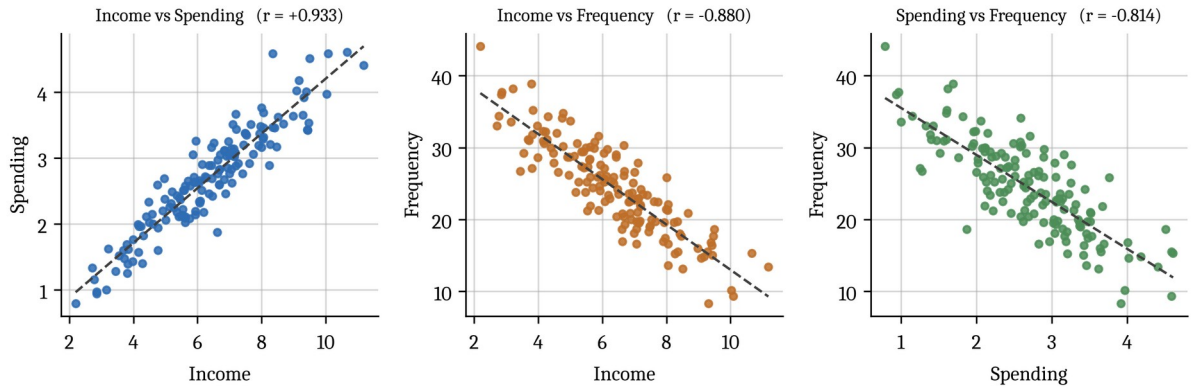
Table 5 - Relationship of every Feature Pair

S.No	Feature Pair	Covariance	Correlation r	Direction	Strength
1	Income – Spending	1.3169	+0.933	Positive	Very strong
2	Income – Frequency	-9.9761	-0.880	Negative	Very strong
3	Spending – Frequency	-4.1199	-0.814	Negative	Very strong

Table 6 - Verification and Summary of the Analysis

Parameter	Value
Number of customers (n)	150
Order of the data matrix X	150 × 3
Order of the covariance matrix S	3 × 3
Mean vector	[6.348, 2.686, 24.495]
Column means of the deviation matrix	[0.000, 0.000, 0.000]
Variance of Income	3.1645
Variance of Spending	0.6300
Variance of Frequency	40.6544
Manual S equals np.cov(X)	Verified
Manual R equals pandas .corr()	Verified
Strongest positive relationship	Income – Spending (r = +0.933)
Strongest negative relationship	Income – Frequency (r = -0.880)

Plot 1 - Pairwise Scatter Plots with the Fitted Trend Lines



Plot 2 - Covariance Map and Correlation Map

